

STEVEN WANG

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SKILLS

CAD & Simulations: CATIA V5, SOLIDWORKS, AutoCAD, STAR-CCM+, Ansys Mechanical, GD&T, Tolerance Stack-up
Testing & Validation: Vibration & durability fixtures, torque testing, thermal cycling, tensile & burst, root-cause analysis
Programming & Software: MATLAB/Simulink, Python, C++, Java, HTML/CSS, Excel/Office 365
Manufacturing: Composites, CNC, Lathe/Mill, 3D Printing, Sheet Metal, Waterjet, Laser Cutting, DFM/DFMA

EXPERIENCE

Schaeffler Motion Technologies

May 2026 – Aug. 2026

Mechanical Design Intern

Chatham, ON

- Developed torque-test fixtures in **CATIA V5** for testing from concept to fabrication, measuring **87%** gearbox efficiency
- Modified complex OEM parts containing **130+** components using gearboxes, enclosures, piping, and frameworks
- Root-caused fixture mounting-angle errors via **tolerance stack-up** and **DFM**, cutting rework **50%** and downtime **16+** hrs
- Drafted **40+** engineering drawings using **GD&T** for fixtures, framing, and GA's for manufacturing and documentation
- Authored the control test procedure for purge pump durability tests, documenting each fixture setup with photos

UW Formula Electric FSAE

Sept. 2024 – Present

Aerodynamics Design Lead, Nosecone & Front Wing

Waterloo, ON

- Lead nosecone and front wing design for a **20-member** aero subteam, packaging sensors and enclosures with chassis
- Redesigned the nosecone and whisker package in **SOLIDWORKS** for **2027** rules, reducing front wing turbulence by **15%**
- Built quarter-car **CFD** simulations in **STAR-CCM+** with a repeatable meshing and **CAD** exchange pipeline
- Swept Whisker span, angle of attack, chord, and camber over **1600+** **CFD** solver iterations, reaching **13%** wake reduction
- Defined core mechanisms for nosecone and whisker mounting for electrical, driveline, and active-aero integration.

UW Formula Electric FSAE

Sept. 2024 – Dec. 2025

Chassis & Mechanical Design Member

Waterloo, ON

- Designed TSSI, dashboard, and electrical enclosures in **SOLIDWORKS**, validating mounts in **Ansys** to **1.5** safety factor
- Built alignment jigs for enclosure mounts, eliminating misalignment on chassis assembly and saving a week of rework
- Manufactured carbon fibre aero and structural parts over **6+** wet layup and resin infusion cycles, mold prep to finish
- Supported the **TSSI** shutdown circuit design with the electrical subteam and managed its **CAD** and document library

Veritas/HDT Automotive Solutions

Sept. 2025 – Dec. 2025

Fluid System and Cooling Loop Testing Intern

Ayr, ON

- Validated **500+** fuel, AC, and transmission lines for OEM manufacturers via thermal, tensile, and burst tests
- Built **6** ball-joint fixture setups and updated **SOLIDWORKS** models, keeping traceability for **15** test samples
- Tracked **13** thermal chamber programs over **12** weeks, analyzing and plotting results in **Excel** for R&D and QA

PROJECTS

Parallel vs Series Cooling Loop Study | *Thermal-Fluid Analysis*

[Link](#) 

- Authored an R&D paper comparing parallel vs series cooling loops using hand calcs and ϵ -NTU radiator modeling
- Recommended a dual series loop with single fans on smaller radiators, increasing flow rate per loop by up to **25%**
- Highlighted improved fan control strategies used for **affinity laws** and **PWM** control, avoiding parallel-fan instability
- Compared total cumulative energy consumption between a dual and single series loop, estimating **60%** reduced power

EV Drive Cycle Model | *MATLAB, Simulink*

June 2026 – Present

- Building a backward-facing **EV** powertrain model in **MATLAB/Simulink** to predict energy use over **EPA** drive cycles
- Modeling road load, a loss-derived motor efficiency map, and inverter losses to capture energy use per cycle
- Estimating battery state of charge with **Coulomb counting** to validate the model against **EPA** cycle results
- Quantifying motor, inverter, and battery **I²R** losses per cycle to size cooling system heat rejection

EDUCATION

University of Waterloo

2024 – 2029

Candidate for Bachelor of Applied Science in Mechanical Engineering

Waterloo, ON